

Compliance & Management Hub (POC)

Combined Project Document — Requirements + Feature Specification

Company: SlashDr

Type: Intern Project (Full-Stack — Backend + Frontend)

Duration: 4 weeks

Status: Draft — pending review by Product/Compliance lead

1. Background & Problem Statement

SlashDr serves clinics and nursing homes that are required to maintain two categories of records for regulatory and legal purposes, both of which are currently handled manually (paper/Excel) by most customers:

- **Patient Informed Consent:** Required for procedures such as surgery, anesthesia, and blood transfusion, per ICMR ethical guidelines and standard medical-legal practice. Paper consent forms are prone to loss, illegible signatures, and are hard to retrieve during audits or disputes.
- **Clinic-Level Regulatory Compliance:** Nursing homes must maintain valid licenses/registrations (e.g., Clinical Establishment registration, Fire NOC, Biomedical Waste authorization, PC-PNDT registration where applicable). Missed renewals carry real legal and financial risk, with no current system to track expiry across a clinic or clinic group.

Both problems share the same underlying need: capture a document/record, attach an expiry or time-bound validity, and alert relevant people before it lapses. This project builds a single module that solves both.

2. Goals

- Digitize patient consent capture, tied to a specific visit/procedure, with e-signature.
- Digitize clinic license/registration tracking, with expiry alerts.
- Provide searchable registers for both, so staff can retrieve records instantly during an audit or inspection.
- Build this as a POC in an isolated module/sandbox — not wired into production data — for evaluation before any go-live decision.

3. Non-Goals (Out of Scope for this POC)

- Legally binding/notarized e-signatures (a basic captured signature image is sufficient for POC).
- Integration with government portals (e.g., auto-submission to State Pollution Control Board, PC-PNDT authority). Export in the correct format is in scope; automated submission is not.
- Billing or payment-related logic.
- Multi-language support (English only for POC).

4. Users & Roles

Role	Access Description
Nursing/OP Staff	Can capture patient consent during a visit. Cannot view clinic compliance dashboard.
Doctor	Can view/capture consent linked to their patients. Read-only on clinic compliance.
Clinic Admin	Full access to clinic's own compliance tracker (licenses) and consent register.
Super Admin (SlashDr internal)	Cross-clinic view — compliance status across all clinics in a group.

5. Feature Specification

Module A: Patient Consent Register

A.1 — Consent Template Library: Predefined, procedure-specific consent form templates that staff select from rather than writing consent text from scratch each time.

Field	Type	Required	Notes
Template Name	Text	Yes	e.g., "Surgical Procedure Consent"
Procedure Type	Dropdown	Yes	Surgical / Anesthesia / Blood Transfusion / Other
Form Body	Rich text w/ placeholders	Yes	Static legal/clinical text + dynamic fields
Requires Witness Signature	Boolean	Yes	Some consents legally require a witness
Active/Inactive	Boolean	Yes	Inactive templates can't be selected for new consents

Business Rules: Only Clinic Admin / Super Admin can create or edit templates. Editing a template does not retroactively change already-signed records (each signed record stores a frozen copy of the form text at signing time). POC ships with 2–3 templates only.

A.2 — Consent Capture Flow: Triggered from patient visit screen ("Add Consent" → pick template). Includes auto-filled demographic data, signature canvas pad, witness validation, and immutable record creation (status: active). Void/re-create mechanics apply for errors.

A.3 — Consent Register (Search & View): Filters by patient name/ID, date range, procedure type, status, and creator. Export supports single PDF or filtered CSV registers.

A.4 — Time-Bound Consent Expiry (stretch): Optional `valid_until` attribute; background job auto-flips status to expired with visual warnings for staff.

Module B: Clinic Compliance Tracker

B.1 — License/Registration Entry

Field	Type	Required	Notes
Clinic	Dropdown	Yes	Auto-set for Clinic Admin
License Type	Dropdown	Yes	Clinical Establishment / Fire NOC / Biomedical Waste / PC-PNDT / Other
License Number	Text	Yes	Unique identifier for document
Issuing Authority	Text	Yes	Governing body

Field	Type	Required	Notes
Issue Date / Expiry Date	Date	Yes	Expiry must be after Issue Date
Uploaded Document	File (PDF/image)	Yes	Verification attachment

B.2 — Expiry Status & Alerts

Days to Expiry	Status	Indicator Color
> 30 days	Valid	Green
8–30 days	Renewal Due Soon	Amber
0–7 days	Urgent	Red
< 0 days	Expired	EXPIRED

B.3 & B.4 — Renewal Flow & Dashboard: Admin clicks "Renew" on expiring records to create linked versions via `renewed_from_id`. Dashboards offer single-clinic cards/tables and Super Admin cross-clinic rollups.

Module C: Shared / Cross-Cutting

- **C.1 Secure Document Storage:** Access-controlled signed URLs, restricted types (PDF/JPG/PNG), enforced payload size limits.
- **C.2 Audit Trail:** Immutable logs tracking all administrative/clinical mutations with metadata JSON.
- **C.3 RBAC Matrix:** Server-side authority boundaries strictly verified for every endpoint call.

6. Data Model (Draft)

```
documents
  id, entity_type (patient|clinic), entity_id, category, file_url,
  issued_date, expiry_date, status, uploaded_by, created_at

consent_templates
  id, name, procedure_type, form_fields (json), requires_witness (bool)

consent_records
  id, template_id, patient_id, visit_id, filled_data (json),
  patient_signature_url, witness_signature_url, captured_by,
  captured_at, valid_until (nullable), status (active|void)

clinic_licenses
  id, clinic_id, license_type, license_number, issuing_authority,
  issue_date, expiry_date, document_url, status, renewed_from_id (nullable),
  created_at, updated_at

audit_log
  id, user_id, action_type, entity_type, entity_id, timestamp, meta (json)
```

7. API Endpoints (Draft)

- **Consent:** GET `/consent-templates` | POST `/consent-records` | GET `/consent-records?`
`patient_id=&date_from=&date_to=&procedure_type=` | GET `/consent-records/:id/export` (PDF)

- **Compliance:** POST /clinic-licenses | GET /clinic-licenses?clinic_id= | GET /clinic-licenses/:id | POST /clinic-licenses/:id/renew | GET /admin/compliance-summary
- **Shared:** POST /documents/upload | GET /audit-log?entity_type=&entity_id=

8. Non-Functional Requirements & 9. Week-by-Week Plan

Category	Requirement Specification
Security	No document/signature URL guessable or publicly accessible without auth.
Performance	Register search returns results in <2s for up to ~10,000 records.
Reliability	Consent records immutable once signed — no update endpoint may edit content.
Auditability	Every write action traceable to a user and timestamp.
Usability	Consent capture completable in under 2 minutes with minimal training.

Week	Focus Area
Week 1	Data model, migrations, file upload/storage setup, core backend scaffolding.
Week 2	Backend APIs — consent capture + license CRUD, expiry background job, RBAC.
Week 3	Frontend — consent capture flow, compliance dashboard, search/register views.
Week 4	Notifications (30/15/7-day alerts), PDF export, audit trail, edge-case QA.

10. Acceptance Criteria & 11. Risks

- [] Staff can capture a consent form with signature, linked to a patient visit, in < 2 minutes
- [] Consent records are searchable and exportable as PDF
- [] Admin can add a clinic license and see correct expiry status (Green/Amber/Red)
- [] Notifications fire correctly at 30/15/7 days before expiry
- [] Super Admin dashboard correctly aggregates status across multiple clinics
- [] A staff user cannot access another clinic's compliance data

Risk Factor	Mitigation Strategy
Consent form fields mismatch actual requirements	Get sign-off from clinical/compliance reviewer before Week 3 frontend work.
File storage exposes sensitive documents	Use signed URLs / access-controlled storage from Day 1.
Scope creep on types/templates	Lock initial list in Week 1; extra items deferred to v2.